YTL*AI LABS Z.AI

BUSINESS PROPOSAL

UMHackathon 2026

umhackathon@um.edu.my

UM HACKATHON 2026 • FINAL ROUND

YTL*AI LABS Z.AI

RAILMIND

Real-Time AI-Driven LRT Management Intelligence Network & Dispatch

"Malaysia's first AI-native train dispatch system : Built to save time, reduce cost, and generate strategic data for the nation."

Submitted to: Universiti Malaya Hackathon 2026 : Final Round Judging Panel
Date: 2nd May 2026

1. Executive Summary

RAILMIND is an AI-driven operational intelligence platform purpose-built to modernize Malaysia's urban rail network. Currently, Rapid KL's LRT frequency adjustments are heavily manual and reactive, leading to severe overcrowding during peak hours, monsoon flash floods, and major stadium events.

The purpose of **RAILMIND** is to transition Prasarana from manual "calls" to automated, data-driven dispatching. By integrating real-time faregate data, weather sensors, and event calendars, our system optimizes train headways to reduce wait times and operational waste. The market for rail digitalization in Malaysia is a critical opportunity, as urban congestion currently drains billions from the national productivity. Our competitive advantage lies in our localized AI model that understands Malaysian environmental variables (like flash floods) better than generic foreign systems. The business model focuses on cost-avoidance for the government and generating high-value operational data that serves as a strategic bargaining asset for future projects like the Penang Mutiara Line.

The impact is threefold:

- Economic: A 15% average commute time reduction translates to RM 2.1 billion annually in recovered labour productivity.
- Operational: Automation of frequency management reduces costly human error and lowers energy expenditure. Three new AI-powered input modules - a CCTV-based crowd density scanner, a concert poster / event ticket OCR parser, and a carbon emission tracker - extend RAILMIND's sensing capability beyond faregate data alone, providing richer real-time inputs and measurable carbon tax savings under Malaysia's 2026 Carbon Tax Act.
- Strategic: Generates a proprietary operational dataset to serve as a sovereign bargaining asset in negotiations with international contractors.

2. Product Overview / Statement

Key Features:

- **Dynamic Frequency Engine:** Automates train dispatch intervals based on real-time station congestion and faregate "tap-in" velocity.
- **Weather-Responsive Logic:** Triggers pre-defined service adjustments during MetMalaysia-verified flash flood warnings.

- **Event Intelligence Module:** Allows operators to upload a concert ticket or poster. Powered by Z.AI's GLM multimodal model, it detects date, time, and estimates concert-goers from past data to predict frequency and load factor.
- **Strategic Data Layer:** Aggregates high-resolution ridership and energy-usage data into a "Data Package" for infrastructure negotiations.
- **Predictive Maintenance Dashboard:** Monitors track and rolling stock health to move away from rigid, calendar-based maintenance schedules.
- **CCTV Vision Intelligence Module:** Operators or automated agents can upload a CCTV snapshot from any station camera feed. A computer-vision model (YOLOv8-based object detection) analyses the image to estimate the real-time headcount on the platform and concourse. This crowd density estimate is fed directly into the Dynamic Frequency Engine as a live signal, supplementing faregate tap-in counts with a visual ground-truth check. In stations where faregate data lags by 2–3 minutes (e.g., during gate malfunctions or staffed entry points), the CCTV Vision Module provides an immediate, camera-derived passenger density input, ensuring dispatch decisions are never made blind.
- **Event Intelligence Module - Concert Poster & Ticket Scanner:** A multi-modal input module that allows OCC operators or an automated web-scraping agent to upload an image of a concert poster, event flyer, or printed ticket. The system uses Optical Character Recognition (OCR) combined with a large language model (LLM) parser to extract structured event metadata: venue, date, start time, expected attendance tier (from venue capacity, e.g., Stadium Merdeka = 100,000 pax; Axiata Arena = 13,000 pax; Zepp KL = 2,500 pax). This metadata is cross-referenced against the nearest LRT station catchment zone. RAILMIND then automatically generates a pre-event service adjustment plan, projecting the estimated surge demand window (typically T-90 minutes to T+60 minutes post-event), recommended frequency uplift (e.g., headway compressed from 5 min to 2.5 min), and the corresponding load factor curve. The plan is presented to the duty OCC manager for one-click approval, replacing the current ad-hoc, memory-dependent process for major events.
- **Real-Time Broadcast NLP Agent:** Scans live news outlets and YouTube summaries. Analyses anchor speech to estimate passenger boarding rates and prepare for external incidents.

Technology Used:

- **Cloud-Native Microservices:** Built for high availability and seamless integration with existing rail signaling PLCs.

- **Demand Forecasting (ML):** Uses Long Short-Term Memory (LSTM) networks to predict passenger surges before they occur.
- **Open API Architecture:** Allows real-time ingestion of external data from weather (MetMalaysia) and event ticketing platforms (Ticket2u).
- **Computer Vision (YOLOv8):** Powers the CCTV Vision Intelligence Module. Accepts JPEG/PNG image uploads from existing station CCTV feeds. The model detects and counts human silhouettes, classifies density zones (sparse / moderate / crowded / critical), and returns an estimated headcount with a confidence interval within 3 seconds of image upload. Operates alongside faregate data as a redundant, visual ground-truth sensor.
- **OCR + LLM Event Parser:** Powers the Concert Poster / Ticket Scanner module. Uses Tesseract OCR for text extraction from uploaded poster or ticket images, followed by an LLM (GPT-4o-mini or locally-hosted Llama 3) to parse unstructured text into structured JSON: {venue, date, start_time, capacity_tier, nearest_lrt_station}. The parsed event is automatically queued in RAILMIND's Event Calendar and triggers a pre-event dispatch plan generation 7 days in advance and a real-time frequency uplift 90 minutes before event start.
- **Carbon Emission Tracker (TNB Grid Factor Integration):** A real-time carbon accounting sub-module that multiplies the traction energy saved per optimised trip against TNB's published grid emission factor (0.571 kg CO_2e/kWh, 2024). The dashboard displays a live "carbon saved today" counter and a monthly carbon liability projection for Prasarana's ESG reporting and Carbon Tax Act compliance filings.

2.1 The Status Quo: Manual Operations

Step	Current Manual Process	Key Failure Mode
1	Station master observes platform crowding visually	Subjective perception varies by individual
2	Calls Operations Control Centre (OCC) via telephone	Communication lag of 5–15 minutes under load
3	OCC coordinator assesses network-wide load manually	No unified data view; relies on heuristics
4	Headway adjustment instruction issued to drivers	Instruction propagates slowly across the line
5	Effect monitored manually; cycle restarts	By the time adjustment takes effect, demand has shifted

3. Market Research & Market Dynamics

3.1 Problem Landscape

Current Malaysian rail operations are constrained by a 10–15 minute manual latency caused by verbal communication between station masters and control centers, a lack of unified data where weather and event systems operate in silos, and a heavy foreign dependency on proprietary signaling "black boxes" that prevents Malaysia from developing the local intellectual property needed to optimize its own infrastructure.

3.2 Market Niche

RAILMIND serves Prasarana Malaysia Berhad and MRT Corp as primary users, providing a more reliable commute for 1.1 million daily Klang Valley residents while equipping local maintenance teams with the digital diagnostic tools necessary to manage aging rolling stock. Beyond Malaysia, the platform is designed to scale as a "Digital Leapfrog" solution for underdeveloped countries—specifically within ASEAN, South Asia, and Africa, that still rely on manual dispatching and legacy rail systems. By providing a software-driven AI overlay, RAILMIND allows these nations to bypass the multi-billion dollar costs of hardware-heavy foreign automation, reclaiming lost national productivity and establishing Data Sovereignty over their own transport infrastructure.

3.3 Market Sizing

Market Tier	Definition	Estimated Value (2026)
TAM	Real-time API connectors	Faregate tap-ins, CCTV, MetMalaysia
SAM	LSTM neural network	30-min frequency forecasts
SOM	Rule-based constraint solver	Optimal headway recommendations

3.4 Market Dynamics & Placement:

- **Infrastructure Momentum:** Malaysia is currently undergoing a massive rail expansion, with projects like the ECRL and the Penang Mutiara LRT creating an urgent demand for localized operational software. This momentum allows **RAILMIND** to enter the market as a primary integration partner rather than a secondary add-on.

- **National Data Sovereignty:** Governments are increasingly wary of "Foreign Dependency," where rail signaling systems act as proprietary "black boxes" owned by international contractors. **RAILMIND** capitalizes on the shift toward "Data Sovereignty," providing Malaysia, and other emerging nations, with a strategic dataset to use as a bargaining asset in infrastructure negotiations.

- **The "Digital Leapfrog" Trend:** There is a global shift toward using AI and software-based overlays to modernize legacy infrastructure. For underdeveloped countries still utilizing manual dispatching systems, **RAILMIND** enables a "digital leapfrog",allowing them to

achieve modern efficiency and recover lost GDP productivity without the multi-billion dollar price tag of hardware-heavy foreign automation.

- **Productivity Pressure:** Mounting public and government pressure to solve "First-and-Last Mile" gaps and peak-hour overcrowding has turned rail efficiency into a top-tier political and economic priority. By framing rail operations as a "GDP Catalyst," **RAILMIND** positions itself as an essential tool for national economic recovery rather than just an operational utility.

4. Value Proposition & Strategic Power

RAILMIND is redefining the economics of Malaysian transport by converting rail operations into a **GDP Catalyst**.

- **Economic Recovery:** By reducing average commute times by 15%, we unlock millions of "man-hours," directly contributing to Malaysia's domestic GDP.

- **Strategic Bargaining Asset:** During negotiations with international partners (such as China for the ECRL or Mutiara Line), Malaysia can use **RAILMIND's** data to prove operational efficiencies. This allows the government to lower "Package" costs by challenging the inflated complexity assumptions of foreign builders.

- **Cost Efficiency:** Automation eliminates human error in dispatch and reduces traction energy costs by 25% during off-peak hours by optimizing empty train runs.

- **Carbon Emission Reduction & Carbon Tax Savings:** RAILMIND directly reduces the total energy consumed per vehicle-kilometer by dynamically optimizing train frequencies during low-demand periods, which not only lowers grid-sourced carbon emissions but also significantly strengthens Prasarana's **Environmental, Social, and Governance (ESG)** profile. By aligning with Malaysia's Carbon Tax framework—which imposes a levy of RM 50–80 per tonne of CO_2e—RAILMIND is projected to eliminate approximately 8,850 tonnes of CO_2e annually on the Kelana Jaya Line alone, translating to RM 443,000–RM 708,000 in avoided tax liability. Network-wide, these savings scale to an estimated **RM 2.2–3.5 million per year**, providing a quantifiable economic benefit while fulfilling ESG reporting requirements and demonstrating a commitment to national sustainability goals under the 2026 Climate Change Act

5. Competitor Analysis (Porter's Five Forces)

Force	Intensity	Assessment
Competitive Rivalry	Medium	Foreign giants (Siemens, Thales) are present but lack the localized "flash flood/event" logic specific to Malaysia.
Threat of New Entrants	Low	High technical barriers and safety-critical compliance make entry difficult for general AI startups.
Bargaining Power of Buyers	High	Prasarana/MOT are the sole buyers; our value is in cost-saving, not high pricing.
Bargaining Power of Suppliers	Medium	Depends on access to government faregate data, which is mitigated by our "National Data Sovereignty" pitch.
Threat of Substitutes	Low	Traditional manual dispatching is the only alternative, but it is increasingly unviable for modern demand.

5.2 The Negotiation Playbook

RAILMIND establishes Malaysia's first rail data sovereignty framework. This dataset allows negotiators to:

- Challenge inflated 'operational complexity' assumptions in contractor pricing.
- Demonstrate that Malaysia's AI can replace proprietary foreign control systems.
- Set contractual KPIs backed by real benchmarks, reducing variation in order risk.

6. Business Model Canvas

6.1 Customer Segmentation

1. **Domestic Transit Operators:** Prasarana and MRT Corp utilize the platform to transition from manual, reactive dispatching to automated, data-driven frequency management.
2. **Global Rail Authorities:** State-owned operators in underdeveloped markets (ASEAN, Africa) use the system as a "Digital Leapfrog" to modernize manual tracks without hardware-heavy debt
3. **Strategic Negotiators:** The Ministry of Transport (MOT) and EPU leverage proprietary "Sovereign Data Packages" to challenge inflated contractor pricing for new projects like the Mutiara Line.

6.2 Revenue Stream:

- **Primary:** SaaS Licensing fee for the RAILMIND platform across the Klang Valley network.
- **Secondary:** Consultancy and Data Package licensing for new rail project bidding cycles (e.g., Mutiara Line).

6.3 Cost Structure

Force	Intensity	Assessment
Competitive Rivalry	Medium	Foreign giants (Siemens, Thales) are present but lack the localized "flash flood/event" logic specific to Malaysia.
Threat of New Entrants	Low	High technical barriers and safety-critical compliance make entry difficult for general AI startups.
Bargaining Power of Buyers	High	Prasarana/MOT are the sole buyers; our value is in cost-saving, not high pricing.
Bargaining Power of Suppliers	Medium	Depends on access to government faregate data, which is mitigated by our "National Data Sovereignty" pitch.
Threat of Substitutes	Low	Traditional manual dispatching is the only alternative, but it is increasingly unviable for modern demand.

6.4 Key Partners

Internal (National Infrastructure)	External (Data & Strategy)
Prasarana Malaysia Berhad & MRT Corp (Operational Access)	**MetMalaysia** (Weather Data Ingestion)
Ministry of Transport (MOT) & EPU (Strategic Alignment)	**Ticket2u / Event Organizers** (Demand Forecasting)
National Cybersecurity Agency (NACSA) (Data Sovereignty Compliance)	**International Rail Consultants** (Global Export Strategy)

6.5 Risk Assessment

Risk Type	Impact	Likelihood	Mitigation
Data Silos (Lack of access to faregate data)	**High**	**Medium**	**Use the "National Data Sovereignty" pitch to align with government security goals.**
Regulatory / Safety Compliance (Critical rail standards)	**High**	**Medium**	**Early proposal submission to SPAD/APAD and partnership with existing signaling contractors.**

Foreign Dependency (Proprietary "Black Box" signaling)	**High**	**High**	**Focus on the "AI Overlay" model that works alongside, rather than replacing, existing PLCs.**
Technical Latency (AI processing lag during peak loads)	**Medium**	**Low**	**Deploy on cloud-native microservices designed for high availability and low-latency response.**
Resistance to Change (Transition from manual "calls")	**Medium**	**Medium**	**Phased implementation: Start with "Decision Support" before moving to full automated dispatch.**

6.6 Future Implementation & Scalability:

RAILMIND is designed for rapid integration into existing Malaysian infrastructure. As a Zero-Hardware' platform, it does not require physical overhaul, only API-level integration with Prasarana's faregate and ATS servers.

6.7 Implementation into Rapid KL

For advanced GoA3 and GoA4 lines, **RAILMIND** acts as a '**Predictive Overlay'**. While these lines are automated, they are reactive. Our code provides the forward-looking telemetry needed to adjust headways before platforms reach capacity. Implementation is achieved through a secure API layer connecting Prasarana's data centers to our predictive microservices.

6.8 Multimodal Expansion (Bus Systems)

The system can be branched into bus networks by analyzing station exit data. By predicting **'discharge waves'** from rail stations, **RAILMIND** can automatically notify feeder bus dispatchers to increase frequency, ensuring a seamless transition across transport modes.

6.9 Global Prospects & Underdeveloped Markets

Our target clients extend beyond Rapid KL to global rail services in underdeveloped countries where manual operations still prevail. The low-cost, zero-hardware nature of **RAILMIND** makes it the ideal digital transformation tool for these markets.

7. Financial Analysis

RAILMIND's success is measured by the reclamation of lost GDP. For every RM 1.00 invested, the Malaysian economy is projected to recover RM 45.00 in productivity.

Metric	Year 1 (Pilot)	Year 2 (Expansion)	Year 3 (National)
Recovered GDP Productivity	RM 92M	RM 418M	RM 860M
Operational Subsidy Savings	RM 18M	RM 55M	RM 140M
Total Economic Value Added	RM 110M	RM 473M	RM 1.0 Billion

7.1 Go-to-Market Strategy :

RAILMIND primarily adopts a **Network-Sequential Integration** strategy, beginning with high-density automated lines before expanding to manual and multimodal systems. The launch will initiate on the Kelana Jaya Line to establish a predictive baseline in a controlled, high-ridership environment.

For the operator and government base, the "**Shadow Mode**" Pilot is the key penetration plan. This allows Prasarana to validate RAILMIND's AI recommendations against manual OCC decisions in real-time without operational risk. Once accuracy is proven, the strategy scales by positioning the platform as a **Sovereign Data Asset** for the Ministry of Transport, providing the necessary leverage to reduce "Package Costs" in upcoming international rail negotiations, such as the Penang Mutiara LRT

7.2 Implementation Plan :

Phase	Action	Timeline	Target
Soft Launch	**Kelana Jaya Line Pilot:** Deploy as a "Shadow Mode" predictive overlay.	Month 1–4	Eliminate 5–15 min verbal lag; 10% peak wait reduction.
Expansion	**Full KV Rollout:** Integrate with MRT1, MRT2, and LRT3 using existing faregate APIs.	Month 4–12	Full predictive synchronization across 5 major lines.
Optimization	**Multimodal & Event AI:** Activate Z.AI GLM ticket/poster scanning and bus feeder sync.	Month 12–18	Reach economic break-even through energy and productivity recovery.
Strategic Layer	**Sovereign Data Package:** Finalize data assets for Penang Mutiara LRT/ECRL negotiations.	Month 18+	5% reduction in international contractor "Package" pricing.

7.3 Revenue Projections(3-Year) :

Unlike traditional startups, RAILMIND's success is measured by the **reclamation of lost GDP** and the reduction of Prasarana's operational deficit.

A. **GDP Catalyst: Recovering Lost Labor Productivity**

The single largest "revenue" source for the nation is the recovery of time. In 2026, the median monthly salary in Kuala Lumpur is approximately **RM 4,064**. Using the **Cost Saving Approach (CSA)**, travel time savings for commuters are valued at the Marginal Product of Labour.

- **Logic:** If RAILMIND reduces the average daily commute by just **6 minutes** (3 mins each way) for 1 million rail passengers, Malaysia recovers **25 million man-hours** annually.
- **Value:** Based on KL's median hourly wage (~RM 23/hr), this contributes an estimated **RM 575 Million annually** back into the productive economy

Metric	Year 1 (Pilot)	Year 2 (Expansion)	Year 3 (National Scale)
Active Rail Lines	1 (Kelana Jaya)	3 (KJ, Ampang, MRT1)	6 (Full KV Network)
Daily Beneficiaries (Commuters)	298,000	795,000	~1,200,000+
Avg. Time Saved per Journey	2.5 Minutes	4.0 Minutes	5.5 Minutes
Recovered GDP Productivity	RM 92M	RM 418M	RM 860M
Operational Subsidy Savings	RM 18M	RM 55M	RM 140M
Carbon Tax Savings (Avoided Liability)	RM 0.4M	RM 2.0M	RM 17.5M (5yr cumulative)
Total Economic Value Added	RM 110M	RM 473M	RM 1.0 Billion
eROI (per RM 1 spent)	RM 11.00	RM 32.50	RM 58.00

7.4 Key Socio-Economic Milestones

- **Month 6 (Efficiency):** Achieve a **15% reduction** in energy intensity per passenger-km on the Kelana Jaya Line pilot.
- **Month 12 (Connectivity):** Increase average passenger throughput by **12%** during monsoon flash flood events through proactive AI-rerouting.
- **Month 18 (Sovereignty):** Finalize the first "Sovereign Rail Data Package" to be used as a primary bargaining asset for the ECRL operational phase negotiations.
- **Month 24 (Sustainability):** Reach a **Mean Kilometre Between Failures (MKBF)** improvement of 30% over 2024 levels, significantly reducing passenger dissatisfaction.

Executive Note: For every **RM 1.00** invested in RAILMIND, the Malaysian economy is projected to recover **RM 45.00** in combined GDP productivity and operational cost savings over a 5-year horizon.